



UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment # 3 CLO Mapping: CLO2

Course Code	MAT 243	Semester	Fall 2025
Course Title	Multivariable Calculus		
Resource Person	Dr. Muhammad Rayees Ahmad		
Assignment Given Date	11-12-2025	Submission Date	15-12-2025
Total Marks	40= 35 (Assignment) + 5 (Quiz-based Viva)		

Name of student	Complete Registration no.	Section	Assignment marks	Viva marks	Total marks

Submission Instructions (Please follow strictly)

- Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the question at the given empty spaces in the assignment PDF file. DO NOT use any other papers. Alternatively, you can only use simple plane white printer pages.

- This is an **INDIVIDUAL ASSIGNMENT**.
- It is mandatory to know the method and procedure of solving both questions. Any question may be asked in the viva.

- Write your complete names and registration numbers on each page of the assignment.
- Follow the deadline. Finish your work one day before, so you could submit in time.

- **LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).**
- **All Plagiarized assignments will be awarded ZERO marks.**

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student signature: _____

Q # 1 [CLO 2] (35 marks):

Consider the following function

$$f(x, y, z) = (y + 3) \ln\left(\frac{yz}{x + 2}\right) + x^2 e^{xyz-3} + \sin(2x + 3y - 4z + 3)$$

- a) [5 marks] Find the partial derivative $f_x(x, y, z)$.
- b) [5 marks] Find the partial derivative $f_y(x, y, z)$.
- c) [5 marks] Find the partial derivative $f_z(x, y, z)$.
- d) [3 marks] Find the partial derivative $f_{xx}(x, y, z)$.
- e) [3 marks] Find the partial derivative $f_{yy}(x, y, z)$.
- f) [3 marks] Find the partial derivative $f_{zz}(x, y, z)$.
- g) [2 marks] Determine the gradient vector $\nabla f(x, y, z)$.
- h) [1 mark] Determine the gradient vector $\nabla f(x, y, z)$ at point $P(1, 1, 1)$.
- i) [2 mark] Find the unit vector of the vector $\vec{u} = \hat{i} + \hat{j} + \hat{k}$.
- j) [4 marks] Find the instantaneous rate of change of the function $f(x, y, z)$ at point $P(1, 1, 1)$ in the direction of the vector $\vec{u} = \hat{i} + \hat{j} + \hat{k}$.
- k) [2 mark] Determine the rate of increase of $f(x, y, z)$ at point $P(1, 1, 1)$ in the direction of the gradient.

Solution:

Name:

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